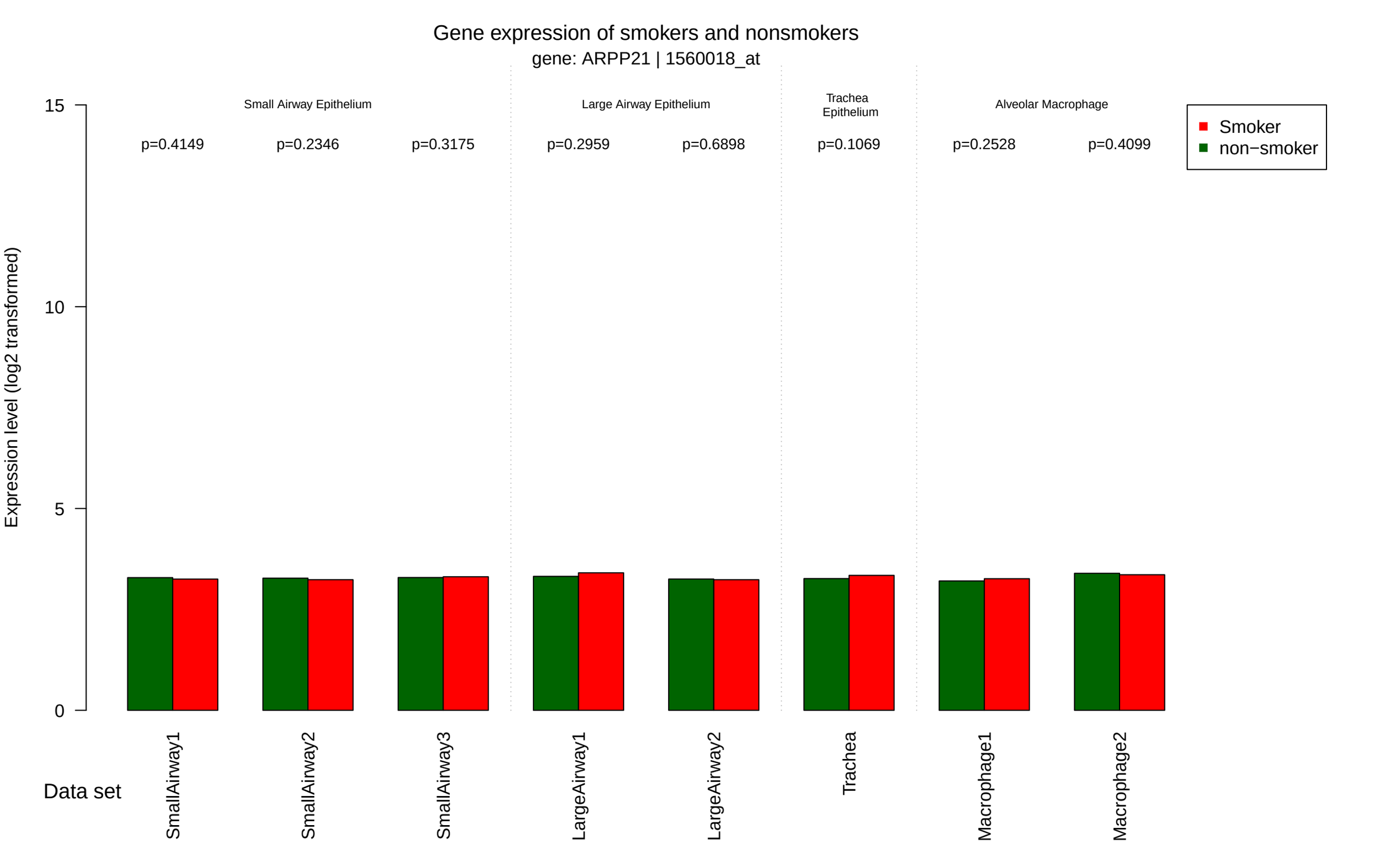
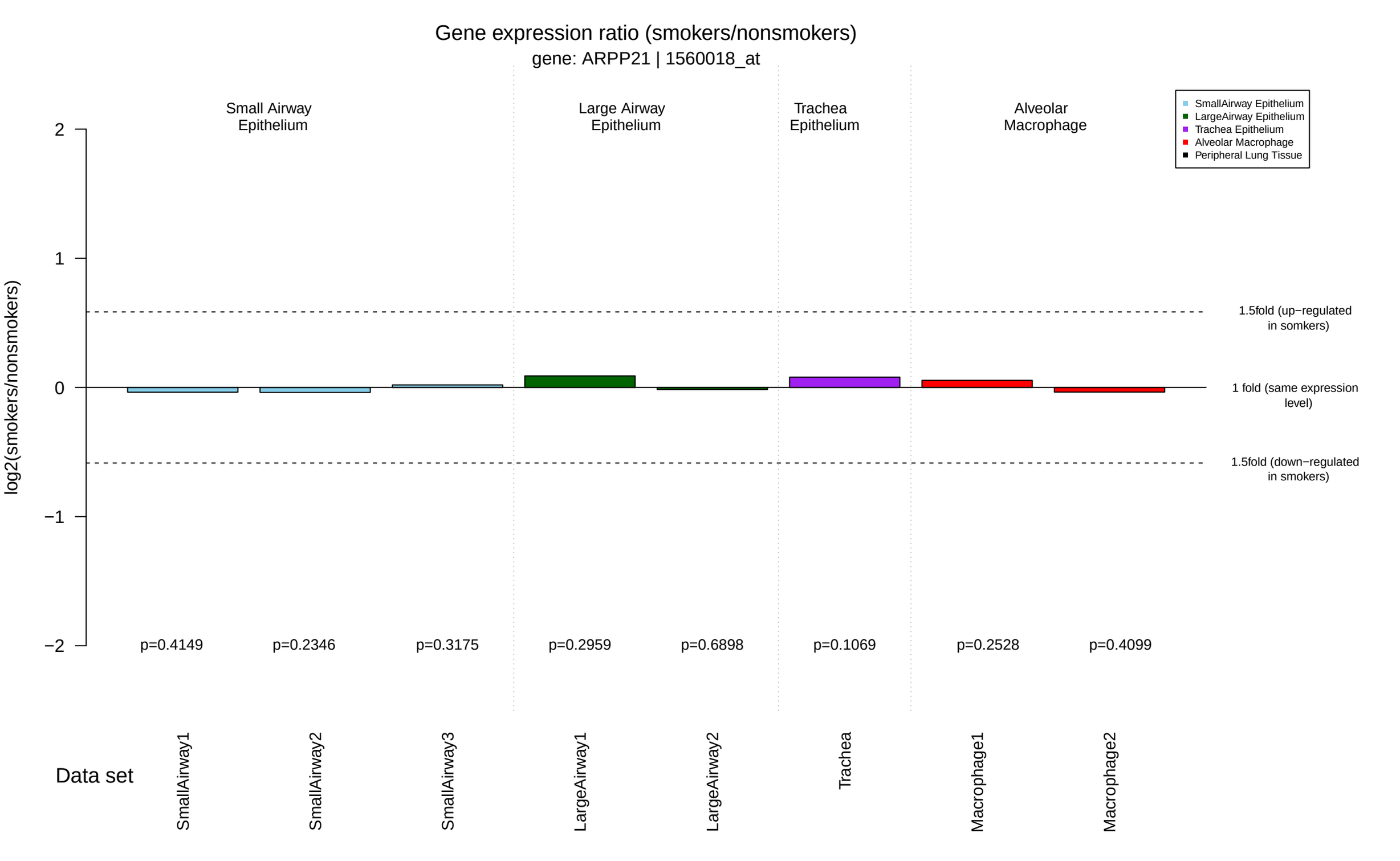
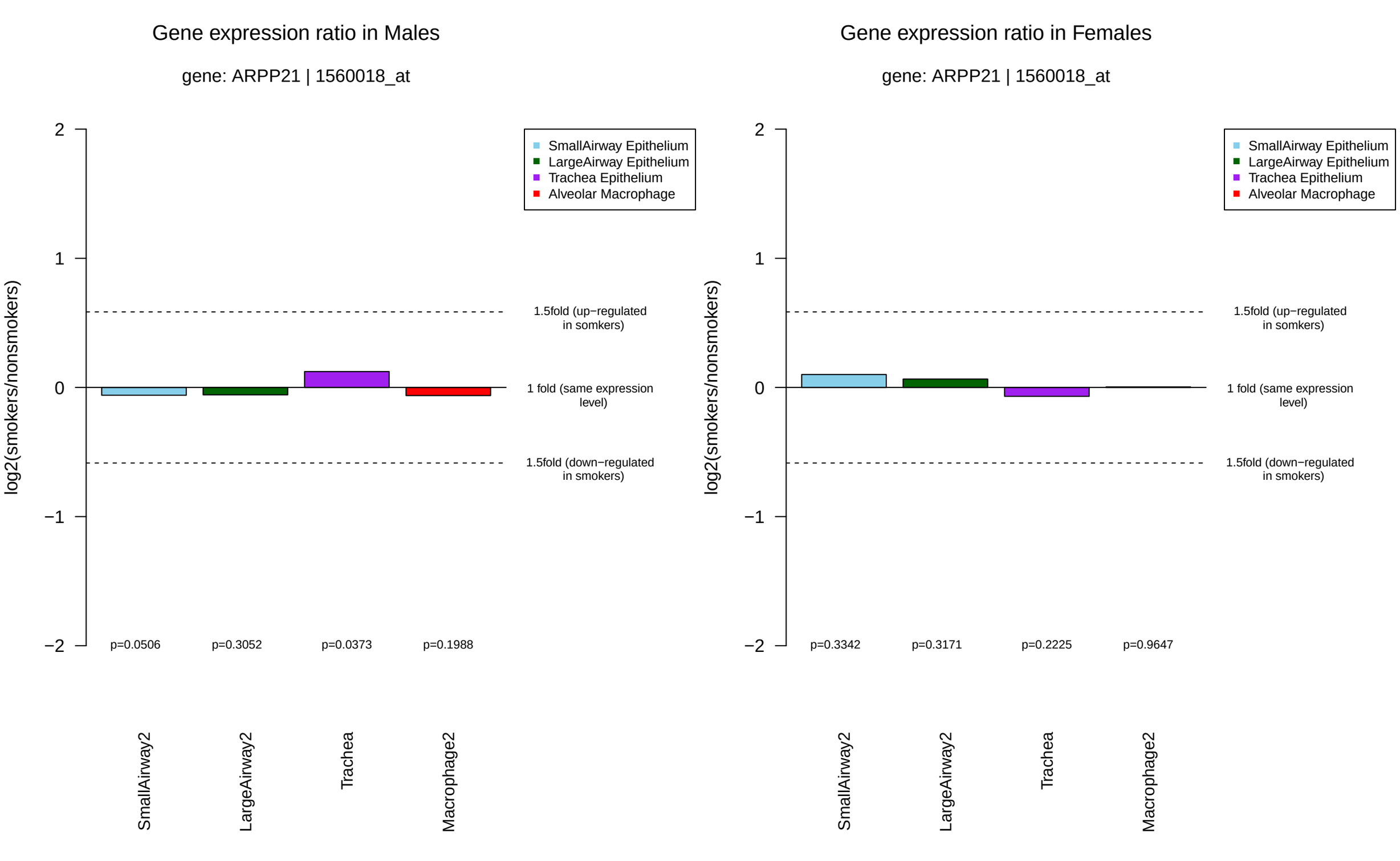
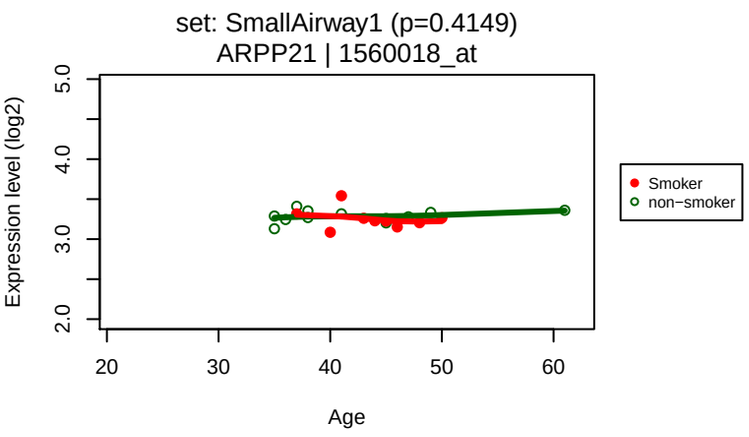
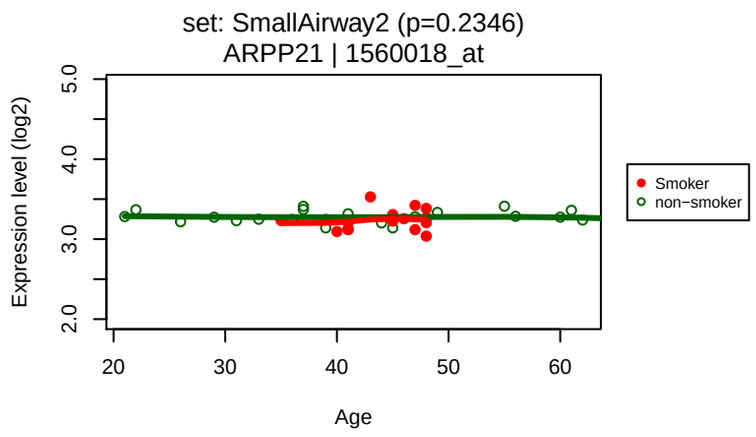
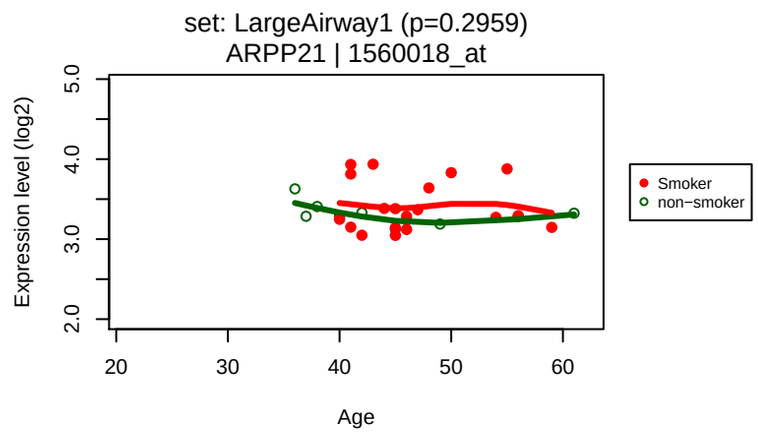

Gene expression of smokers and nonsmokers
gene: ARPP21 | 1560018_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4149
p=0.2346
p=0.3175
p=0.2959
p=0.6898
p=0.1069
p=0.2528
p=0.4099
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

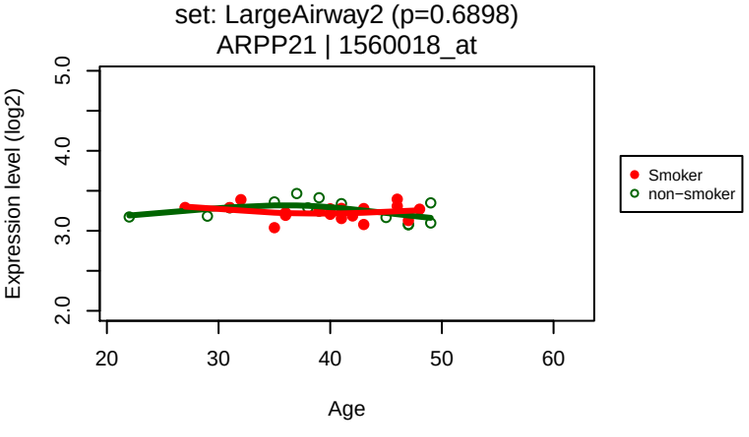

Gene expression ratio (smokers/nonsmokers)
gene: ARPP21 | 1560018_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4149
p=0.2346
p=0.3175
p=0.2959
p=0.6898
p=0.1069
p=0.2528
p=0.4099
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

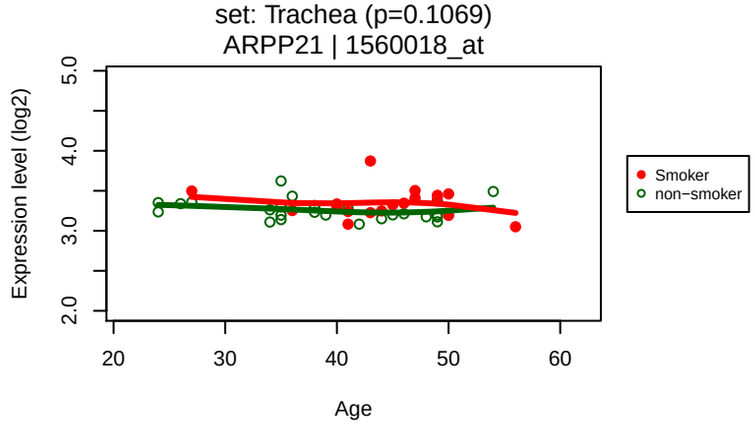

Gene expression ratio in Males
gene: ARPP21 | 1560018_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0506
p=0.3052
p=0.0373
p=0.1988
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ARPP21 | 1560018_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3342
p=0.3171
p=0.2225
p=0.9647
SmallAirway2
LargeAirway2
Trachea
Macrophage2

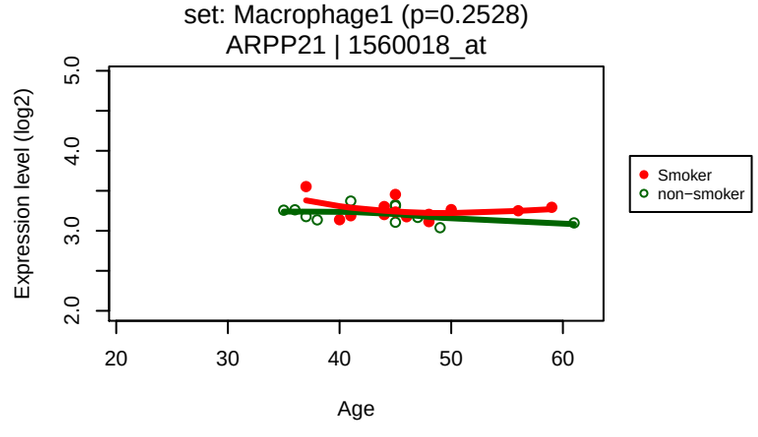

set: SmallAirway1 (p=0.4149)
ARPP21 | 1560018_at
Expression level (log2)
Age
Smoker
non−smoker

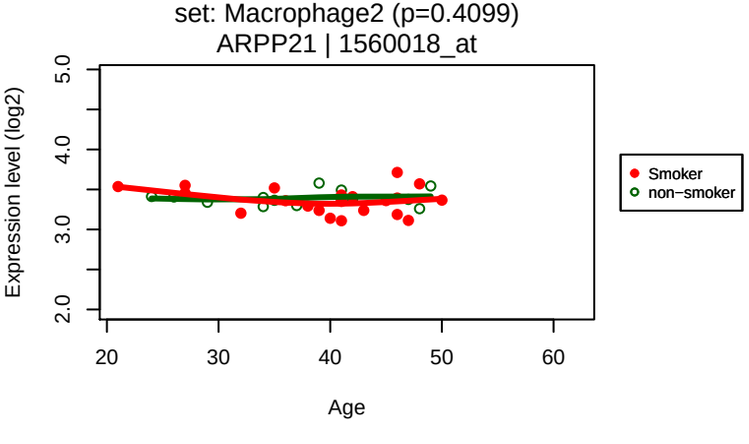

set: SmallAirway2 (p=0.2346)
ARPP21 | 1560018_at
Expression level (log2)
Age
Smoker
non−smoker

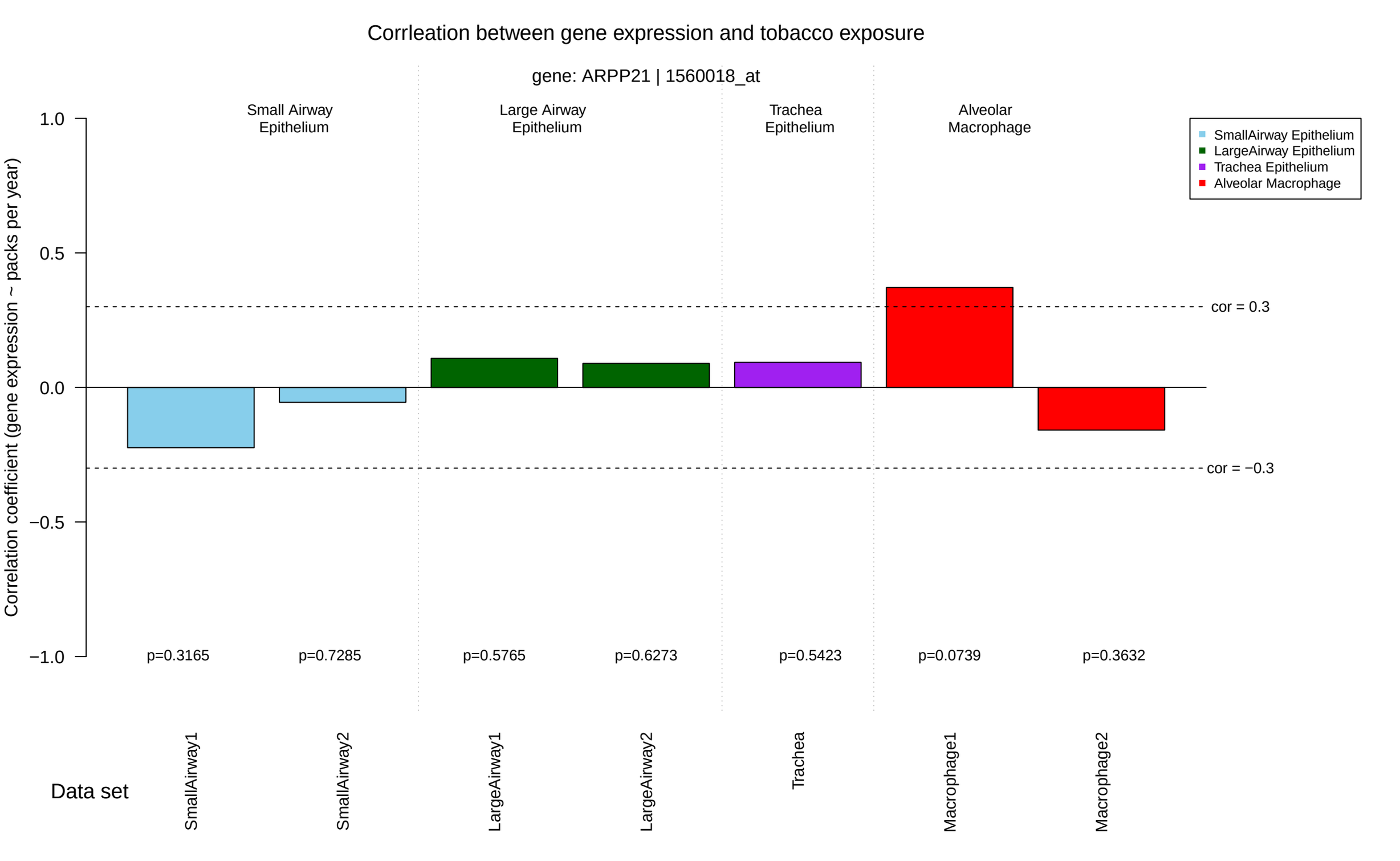

set: LargeAirway1 (p=0.2959)
ARPP21 | 1560018_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6898)
ARPP21 | 1560018_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1069)
ARPP21 | 1560018_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2528)
ARPP21 | 1560018_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4099)
ARPP21 | 1560018_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ARPP21 | 1560018_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3165
p=0.7285
p=0.5765
p=0.6273
p=0.5423
p=0.0739
p=0.3632
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2